Haizhong Zheng

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INTERESTS

Efficient RL for LLMs; Machine Learning System; Agent; ML Security.

My research focuses on building models, algorithms, and systems for scalable and efficient ML. The goal is to bridge the gap between the rapid scaling of models and the slower scaling of hardware and high-quality data. In particular, my recent work has been along three lines: 1) Efficient reinforcement learning algorithms and systems for LLMs and reasoning. 2) Designing and training hardware-aware and system-aware models for fast model inference. 3) Designing algorithms for efficient data selection, augmentation, and human feedback.

WORK EXPERIENCE

Carnegie Mellon University (CMU) Jan. 2025 - Current
Postdoctoral Researcher, InfiniAI Lab; Member of Catalyst Group
Advisor: Prof. Beidi Chen

Lawrence Livermore National Laboratory (LLNL) May. 2023 - Aug. 2023
Research Intern
Mentor: Dr. Bhavya Kailkhura

Amazon Web Service (AWS) May. 2021 - Aug. 2021
Applied Scientist Intern
Mentor: Dr. Wei Zhang and Dr. Qian Cui

EDUCATION

University of Michigan, Ann Arbor 2024
Ph.D., Computer Science
Advisor: Prof. Atul Prakash

Shanghai Jiao Tong University (SJTU) 2015, 2018
B.S. & M.S., Computer Science and Engineering
Advisor: Prof. Haojin Zhu

SELECTED PUBLICATIONS

- ServeLearnBench: How Well Can Agents Self-Improve from Serving Experience?
  Haizhong Zheng, Yizhuo Di, Ranajoy Sadhukhan, Shuowei Jin, Beidi Chen
  Preprint
- Self-Evolving Defense: Continual Security Policy Learning for LLM Agents
  Minh Nhat Le, Nisarga Gondi, Yibo Peng, Ronghao Ni, Limin Jia, Beidi Chen, Haizhong Zheng
  Preprint
- Sparrow: Sparse Rollout for Stable and Efficient Long-context RL of Large Language Models
  Yang Zhou, Ranajoy Sadhukhan, Zhaofeng Sun, Zhuoming Chen, Souvik Kundu, Saket Dingliwal, Sai Muralidhar Jayanthi, Aram Galstyan, Haizhong Zheng, Beidi Chen
  Preprint
- AstraFlow: Dataflow-Oriented Reinforcement Learning for Agentic LLMs
  Haizhong Zheng, Yizhuo Di, Jiahui Wang, Shuowei Jin, Xueshen Liu, Yongji Wu, Z. Morley Mao, Ion Stoica, Jiawei Zhao, Beidi Chen
  NeurIPS 2026

- The Last Human-Written Paper: Agent-Native Research Artifacts
  Jiachen Liu, Jiaxin Pei, Jintao Huang, Chenglei Si, Ao Qu, Xiangru Tang, Runyu Lu, Lichang Chen, Xiaoyan Bai, Haizhong Zheng, et al.
  NeurIPS 2026
- MonarchRT: Efficient Attention for Real-Time Video Generation
  Krish Agarwal, Zhuoming Chen, Cheng Luo, Yongqi Chen, Haizhong Zheng, Xun Huang, Atri Rudra, Beidi Chen
  NeurIPS 2026
- CamoDocs: A Poisoning Attack Against Retrieval-Augmented Language Models Using Camouflaged Documents
  Jaewon Jung, Haizhong Zheng, Hongsun Jang, Jaeyong Song, Beidi Chen, Jinho Lee
  EMNLP 2026
- Beyond Confidence: Test-Time Scaling for Multi-Turn Search Agents via Retrieval Grounding
  Hyunho Kook, Junhyuk So, Tianyu Fu, Haizhong Zheng, Beidi Chen
  EMNLP 2026 Findings
- When "Correct" Is Not Safe: Can We Trust Functionally Correct Patches Generated by Code Agents?
  Yibo Peng*, James Song*, Lei Li*, Xinyu Yang, Mihai Christodorescu, Ravi Mangal, Corina Pasareanu, Haizhong Zheng, Beidi Chen
  ACL 2026
- XRPO: Pushing the limits of GRPO with Targeted Exploration and Exploitation
  Udbhav Bamba*, Minghao Fang*, Yifan Yu, Haizhong Zheng, Fan Lai
  ICML 2026
- Prosperity before Collapse: How Far Can Off-Policy RL Reach with Stale Data on LLMs?
  Haizhong Zheng, Jiawei Zhao, Beidi Chen
  ICLR 2026
- Jackpot: Optimal Budgeted Rejection Sampling for Extreme Actor-Policy Mismatch Reinforcement Learning
  Zhuoming Chen*, Hongyi Liu*, Yang Zhou*, Haizhong Zheng, Beidi Chen
  ICLR 2026
- OPPO: Accelerating PPO-based RLHF via Pipeline Overlap
  Kaizhuo Yan*, Yingjie Yu*, Yifan Yu, Haizhong Zheng, Fan Lai
  ICLR 2026
- EchoScriptor: Automatic Lifelogging Narratives via Activity-Based Audio–Language Model
  Kaylee Yaxuan Li, Xinghao Zhou, Haizhong Zheng, Kang G. Shin, Alanson P. Sample
  CHI 2026
- RLBoost: Harvesting Preemptible Resources for Cost-Efficient Reinforcement Learning on LLMs
  Yongji Wu*, Xueshen Liu*, Haizhong Zheng, Juncheng Gu, Beidi Chen, Z. Morley Mao, Arvind Krishnamurthy, Ion Stoica
  NSDI 2026

- WAVE: Leveraging Architecture Observation for Privacy-Preserving Model Oversight
  Haoxuan Xu*, Chen Gong*, Beijie Liu*, Haizhong Zheng, Beidi Chen, Mengyuan Li
  ASPLOS 2026
- Act Only When It Pays: Efficient Reinforcement Learning for LLM Reasoning via Selective Rollouts
  Haizhong Zheng, Yang Zhou, Brian R. Bartoldson, Bhavya Kailkhura, Fan Lai, Jiawei Zhao, Beidi Chen
  NeurIPS 2025
- Kinetics: Rethinking Test-Time Scaling Laws
  Ranajoy Sadhukhan*, Zhuoming Chen*, Haizhong Zheng, Yang Zhou, Emma Strubell, Beidi Chen
  NeurIPS 2025
- Plato: Plan to Efficiently Decode for Large Language Model Inference
  Shuowei Jin*, Xueshen Liu*, Yongji Wu*, Haizhong Zheng, Qingzhao Zhang, Atul Prakash, Matthew Lentz, Danyang Zhuo, Feng Qian, Z. Morley Mao
  COLM 2025
- Harmful Terms and Where to Find Them: Measuring and Modeling Unfavorable Financial Terms and Conditions in Shopping Websites at Scale
  Elisa Tsai, Neal Mangaokar, Boyuan Zheng, Haizhong Zheng, Atul Prakash
  WWW 2025, *Oral*
- ELFS: Label-Free Coreset Selection with Proxy Training Dynamics
  Haizhong Zheng*, Elisa Tsai*, Yifu Lu, Jiachen Sun, Brian R. Bartoldson, Bhavya Kailkhura, Atul Prakash
  ICLR 2025
- Learn To be Efficient: Build Structured Sparsity in Large Language Models
  Haizhong Zheng, Xiaoyan Bai, Xueshen Liu, Z. Morley Mao, Beidi Chen, Fan Lai, Atul Prakash
  NeurIPS 2024, *Spotlight*
- Leveraging Hierarchical Feature Sharing for Efficient Dataset Condensation
  Haizhong Zheng, Jiachen Sun, Shutong Wu, Bhavya Kailkhura, Zhuoqing Mao, Chaowei Xiao, Atul Prakash
  ECCV 2024
- CALICO: Self-Supervised Camera-LiDAR Contrastive Pre-training for BEV Perception
  Jiachen Sun, Haizhong Zheng, Qingzhao Zhang, Atul Prakash, Z. Morley Mao, Chaowei Xiao
  ICLR 2024
- Coverage-centric Coreset Selection for High Pruning Rates
  Haizhong Zheng, Rui Liu, Fan Lai, Atul Prakash
  ICLR 2023
- Efficient Adversarial Training with Transferable Adversarial Examples
  Haizhong Zheng, Ziqi Zhang, Juncheng Gu, Honglak Lee, Atul Prakash
  CVPR 2020
- Analyzing the Interpretability Robustness of Self-Explaining Models
  Haizhong Zheng, Earlence Fernandes, Atul Prakash
  ICML 2019 Workshop

- Smoke Screener or Straight Shooter: Detecting Elite Sybil Attacks in User-Review Social Networks

  Haizhong Zheng, Minhui Xue, Hao Lu, Shuang Hao, Haojin Zhu, Xiaohui Liang, Keith Ross

  NDSS 2018

(* indicates equal contribution)

GRANT PROPOSALS

I actively contributed to the proposal design, proposal writing, and presentation for the following grants:

Assessing and Improving Safety and Alignment of LLMs — NAIRR Pilot, 2024
GPU budgets — PI: Atul Prakash

Data Efficiency of LLMs Fine-tuning with RLHF — Cisco, 2023
$150*K* — PI: Atul Prakash

Intelligent Assistants for Detecting Social Engineering Scams — OpenAI, 2023
$100*K* — PI: Atul Prakash

TEACHING

Co-Lead Instructor — *Winter 2023*
Secure and Trustworthy Machine Learning — UMich EECS 598

- My responsibilities include designing the course, teaching the lectures, leading discussions, advising course projects, and office hours.
- EECS 598 covers research topics in machine learning security and privacy.
- Course rating: 4.9 out of 5; Instructor rating: 5 out of 5.

Graduate Student Instructor — *Fall 2021*
Data Structures and Algorithms — UMich EECS 281

- My responsibilities include teaching lectures, leading discussions on a weekly lab section, grading, and holding 6hrs of office hours each week.

STUDENT MENTORED

I am fortunate to mentor and co-advise the research of the following students:

- **Ph.D. Students**: Elisa Tsai (Umich CSE), Shuowei Jin (Umich CSE), Ruiyang Zhu (Umich CSE), Xueshen Liu (Umich CSE)
- **Masters**: Tianyu Zhang (→ ByteDance), Tiejin Chen (→ ASU Ph.D.), Shutong Wu (→ UWsic Ph.D.), Dongfang Ling (CMU CS), Yibo Peng (CMU CS)
- **Undergraduates**: Ziqi Zhang (→ Google), Xiaoyan Bai (→ UChicago Ph.D.), Yifu Lu (→ Princeton Ph.D.), Chengshuo Jiang (Umich CSE), James Song (Umich CSE), Lei Li (Peking University), Qilong Feng (NUS)

INVITED TALKS

- Amazon SF AI Lab — Oct. 2025

SERVICE

Conference/Journal Paper Reviewer: ECCV 2020, TPAMI 2020, ICLR 2022-2024, NeurIPS 2022-2025, ICML 2023-2025, AAAI 2024, ICML 2024 ES-FoMo-II Workshop

AWARDS & HONORS

- Rackham Travel Grant: ICML'19 — 2019
- Academic Excellence Scholarship of SJTU — 2015-2017
- Best Presentation Award in A3 Foresight Program 2015 Annual Workshop — 2015
- Academic Excellence Scholarship of SJTU — 2012-2014

- National Olympiad in Informatics in Provinces (NOIP), First Prize 2010

REFERENCES

Prof. Atul Prakash
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Computer Science & Engineering
University of Michigan
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Prof. Beidi Chen
Assistant Professor
Electrical and Computer Engineering
Carnegie Mellon University
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Prof. Chaowei Xiao
Assistant Professor
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cxiao34@wisc.edu

Dr. Bhavya Kailkhura
Principal Research Scientist
Center for Applied Scientific Computing
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kailkhura1@llnl.gov